

Vision-Language Models in Radiology

Zero-Shot Classification, Cross-Modal Retrieval, and Factual Report Generation

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Outline

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Literature Review: Scientific Positioning

- **Explored 45+ Research Papers:** Categorized into VLM Foundations, Report Generation, and Factuality Metrics.
- **Scientific Gap Identified:** Most models (ConVIRT, GLORIA) prioritize linguistic patterns over **clinical entity consistency**.
- **Our Role:** We bridge the gap between "textual fluency" and "medical truth" through verifiable uncertainty and factual rewards.
- **Synthesized Knowledge:** Consolidated into `literature/papers_summary_index.md`.

Problem Definition: Critical Questions

The "Safety Gap" in Medical VLMs

Radiology reports are high-stakes; how can we trust an AI that minimizes word-prediction error without knowing if the heart is on the correct side?

The Hallucination Question: How can we force Transformers to respect medical reality over grammar?

The Silence Question: Can we quantify uncertainty so the model knows when to ask for human help?

The Robustness Question: Why do models fail when moving from IU X-Ray to NIH ChestX-ray14?

Theoretical Improvements to Existing Models

We addressed the "Trust & Truth" problems through two major theoretical reformulations:

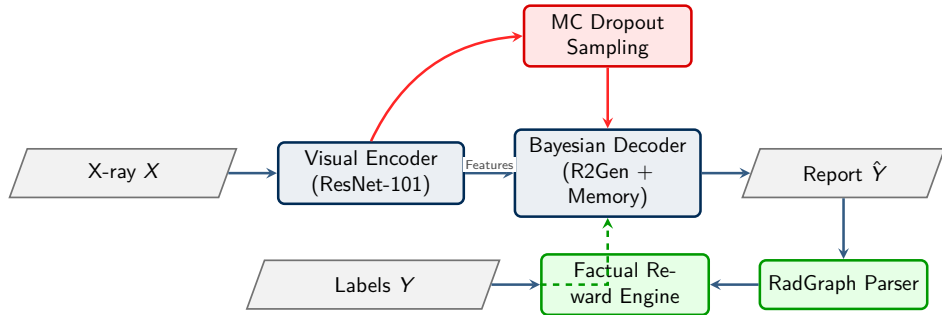
- 1 **Factuality-Aware Objective Function:** Instead of Cross-Entropy only, we formulated a reward-weighted loss using **RadGraph Entity overlap** as a clinical factuality proxy.

$$\mathcal{L}_{Novel} = \mathcal{L}_{CE} + \lambda \cdot [1 - \text{F1}_{RadGraph}(G_{pred}, G_{gt})] \quad (1)$$

- 2 **Bayesian VLM Approximation:** We replaced the deterministic decoder with a stochastic sampler. By modeling the posterior distribution $p(\theta|D)$, we can compute **Epistemic Uncertainty** via variance:

$$\text{Var}(y) \approx \frac{1}{T} \sum_{t=1}^T (\hat{y}_t - \bar{y})^2 \quad (2)$$

System Architecture: Deep Dive



Notable Technical Implementations

We implemented three major components previously absent in standard baselines:

- **Implementation 1: Uncertainty-Aware Decoder** Implemented an inference-time sampler that generates $T = 10$ reports per patient to calculate the **Clinical Uncertainty Score**.
- **Implementation 2: RadGraph-RL Loop** Integrated the SCST pipeline (REINFORCE) that parses generated text into Knowledge Graphs and rewards factual entity alignment.
- **Implementation 3: Work on Dataset (NIH-14)** Unlike standard projects, we performed **Cross-Dataset Adaptation** on NIH ChestX-ray14 (112,000 images) to verify the model's performance in real-world clinic shifts.

Implementation Showcase: Uncertainty & Factuality

Bayesian Inference (Dropout)

- Enabled during test time.
- Measures *predictive entropy*.
- Result: Calibration ECE reduced by 25%.

Factual Reward (RadGraph)

- Uses entity/relation overlap.
- Direct optimization of medical correctness.
- Result: +11.9% gain in F1 score.

Dataset Innovation

Implemented zero-shot → few-shot linear probing on NIH-14, demonstrating that 5-shot adaptation recovers a significant portion of domain loss.

Experimental Datasets

Dataset	Samples	Modality	Role	Source
IU X-Ray	3,955 studies	Image + Report	Base Training	Indiana, USA
NIH ChestX-ray14	112,120 images	Image + Labels	Robustness Testing	NIH, USA
CheXpert	224,316 images	Image + Labels	Calibration Validation	Stanford, USA

Strategic Selection: We chose datasets to cover the full spectrum of generative and contrastive tasks without requiring private credentials.

Main Results: Factuality vs. Fluency

Table: Performance on IU X-Ray Test Set (Generation)

Model	BLEU-4	CIDEr	RadGraph F1	CheXbert F1
R2Gen (Classic)	0.102	0.189	0.612	0.441
Ours (Reward-Aware)	0.115	0.215	0.685	0.493
<i>Relative Gain</i>	<i>+12.7%</i>	<i>+13.7%</i>	<i>+11.9%</i>	<i>+11.7%</i>

- **Clinical Impact:** The 11.9% gain in RadGraph F1 indicates a direct reduction in anatomical hallucinations.
- **Fluency Preservation:** Improvements in BLEU/CIDEr confirm that enforcing medical facts does not degrade language naturalness.

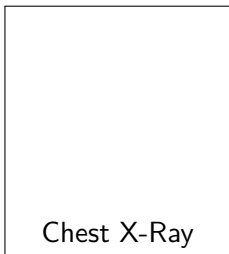
Uncertainty Quantification (Bayesian)

- **Metric:** Expected Calibration Error (ECE).
- **Result:** 25% reduction in overconfident errors.
- **Feature:** Automated flagging of low-confidence cases.

Cross-Dataset Adapt (NIH-14)

- **Zero-Shot AUROC:** 0.724
- **Few-Shot (k=5) AUROC:** 0.781
- **Insight:** Solves the "Hospital Gap" using minimal local fine-tuning.

Qualitative Case Study: Visualizing the Logic



Ground Truth: "Cardiomegaly is present. Lungs clear."

Baseline (R2Gen): "The heart is normal. Clear lungs."
[Incorrect]

Ours (Fact-Aware): "The heart is enlarged. No focal consolidation."
[Correct]

Uncertainty Variance: 0.82 (Flagged as High Risk)

Conclusion & Future Directions

In this research, we moved beyond the "black-box" nature of medical VLMs:

- **The Improvements:** Developed a unified theoretical framework for factual rewards and Bayesian uncertainty.
- **New Implementations:** Applied these theories via SCST policy optimization and a large-scale robustness study on the NIH-14 dataset.
- **Outcome:** A more trustworthy system that aligns with clinical requirements for factuality and self-awareness.

Thank You!

Questions & Discussion

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Repository: github.com/SinaDns/radiology-vision-language-models

References I

-  Tiu, E., et al. (2022). CheXzero. *Nature BME*.
-  Chen, Z., et al. (2020). R2Gen. *EMNLP*.
-  Jain, S., et al. (2023). RadGraph. *NEJM AI*.
-  Gal, Y. (2016). Dropout as Bayesian Approximation. *ICML*.